Hammad Farooq

Computational Scientist, Genomics and Bioinformatics

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AI/ML and computational genomics researcher with a Ph.D. in Bioinformatics. Experienced in developing machine learning, deep learning, and interpretable computational methods for large-scale genomic and biomedical data, with expertise in PyTorch, TensorFlow, multimodal data integration, and GPU/HPC computing.

Education

University of Illinois Chicago July 2026
Ph.D. in Bioinformatics *Chicago, IL, USA*

- **Research Focus**: Artificial Intelligence and Machine Learning for Genomics and Computational Biology.
- **Dissertation**: From Genome Folding to Transcription Factor Rewiring: Decoding Cell Fate Regulation.
- **Advisors**: Constantinos Chronis, Jie Liang

FAST National University of Computer and Emerging Sciences June 2018
Bachelor of Science in Computer Science *Islamabad, Pakistan*

- **Final Year Project**: Predicting the Pathogenicity of Protein-Coding Mutations Using Natural Language Processing.
- **Advisor**: Hammad Naveed

Research & Professional Experience

Johns Hopkins University August 2026 – Present
Malone Postdoctoral Fellow *Baltimore, MD, USA*

- Develop machine learning and deep learning methods for genomic sequence analysis, regulatory variant interpretation, and disease risk prediction.
- Integrate multimodal genomic and genetic data to model gene regulation and connect genetic variation to disease-relevant molecular mechanisms.

University of Illinois Chicago Aug 2019 – July 2026
Graduate Research Assistant *Chicago, IL, USA*

- Developed machine learning and computational models integrating genomic sequence, chromatin accessibility, and 3D genome organization to study gene regulation and predict functional effects of noncoding variants.
- Designed scalable computational pipelines for large-scale multimodal datasets integrating RNA-seq, ATAC-seq, Hi-C, eQTL, GWAS, and single-cell genomic data.
- Applied interpretable deep learning and feature-attribution methods to identify regulatory sequence features and transcription factor motifs underlying context-dependent gene regulation.
- Executed large-scale computational and machine learning workflows on leadership-class HPC and GPU systems using parallel and distributed computing.
- Collaborated with computational, experimental, and clinical researchers on interdisciplinary genomics and precision medicine projects.

FAST National University of Computer and Emerging Sciences Aug 2018 – Aug 2019
Research Assistant – Computational Biology Research Lab *Islamabad, Pakistan*

- Developed machine learning and deep learning models for biomedical applications, including histopathology image analysis and protein structure prediction.
- Applied natural language processing and machine learning to large-scale biomedical and pharmacovigilance data.

Teaching Assistant – Applied Programming *Fall 2018*

- Conducted hands-on C++ programming sessions for graduate students from non-computer-science backgrounds, with a focus on programming fundamentals, problem solving, and debugging.
- Provided guidance on programming assignments and practical implementation of computational concepts.

Automotive Artificial Intelligence Jan 2018 – Feb 2018
AI Developer Intern *Islamabad, Pakistan*

- Developed machine learning models for intrusion detection using high-dimensional cybersecurity data.

- Built data processing, feature engineering, and model evaluation pipelines for real-time anomaly detection.

FAST National University of Computer and Emerging Sciences Fall 2016
Lab Demonstrator – Computer Organization and Assembly Language *Islamabad, Pakistan*

- Led laboratory sessions in computer architecture and assembly language programming, covering instruction execution, processor organization, registers, and low-level programming.
- Guided students in assembly programming, instruction-set simulation, and register-level debugging.

Technical Skills

Artificial Intelligence & Machine Learning: Deep learning, statistical learning, sequence modeling, classification, regression, interpretable AI, feature attribution, and motif discovery; applications in regulatory genomics, variant effect prediction, and biological sequence analysis.

Programming & AI Frameworks: Python, R, C++; PyTorch, TensorFlow, scikit-learn; development of scalable and reproducible machine learning and data analysis pipelines.

GPU & High-Performance Computing: GPU acceleration, parallel and distributed computing, OpenMP, OpenACC, Parsl, and large-scale workflow execution on leadership-class HPC systems including Aurora and Polaris.

Computational Genomics & Multi-Omics: RNA-seq, ATAC-seq, ChIP-seq, Hi-C, single-cell genomics, GWAS, and QTL integration for modeling gene regulation, regulatory variant effects, and enhancer–gene interactions.

Statistical Genetics & Population-Scale Analysis: GWAS and QTL analysis, fine-mapping, variant-set testing, polygenic risk modeling, and analysis of large population cohorts including GTEx and All of Us.

Bioinformatics Software: Scanpy, Seurat, Bioconductor, PLINK2, VCFtools, ChromBPNet, TF-MoDISco, FiNeMo, ChromHMM, Juicer.

Selected Publications

H. Farooq, L. Du, P. Delafrouz, et al. "A New Approach for Discovering Functional Links Connecting Non-Coding Regulatory Variants to Gene Targets." *bioRxiv*, 2024.

L. Du, **H. Farooq**, P. Delafrouz, J. Liang. "Structural basis of differential gene expression at eQTL loci from high-resolution ensemble models of 3D single-cell chromatin conformations." *Bioinformatics*, 41(2): btaf050, 2025.

P. Delafrouz, **H. Farooq**, et al. "Effects of chromatin-lamina attachment on extra-long-range chromatin interactions." *Biophysical Journal*, 2026.

A. Owais, H. Chen, **H. Farooq**, et al. "Gene-gene interactions between a LMNA variant and common polymorphisms drive early-onset atrial fibrillation." *Nature Communications*, 2026.

S. Priyadarshi, K. H. Bhat, E. B. Drake, S. Naiyer, **H. Farooq**, et al. "Igh nuclear speckle association promotes V(D)J recombination." *Nature Communications*, 2026.

K. H. Bhat, S. Priyadarshi, S. Naiyer, X. Qu, **H. Farooq**, et al. "An Igh distal enhancer modulates antigen receptor diversity by determining locus conformation." *Nature Communications*, 14: 1225, 2023.

P. Delafrouz, T. Nesterova, F. E. Harris, **H. Farooq**, et al. "Disentangling polymer confinement from specific-folding interactions reveals the drivers of Escherichia coli chromosome organization." *Nucleic Acids Research*, 54(19): gkag979, 2026.

M. Chen, L. Du, S. Zhao, B. Ye, P. Delafrouz, **H. Farooq**, et al. "ChromPolymerDB: a high-resolution database of single-cell 3D chromatin structures for functional genomics." *Nucleic Acids Research*, 2026.

H. Farooq, J. S. Niaz, S. Fakhar, H. Naveed. "Leveraging digital media data for pharmacovigilance." *AMIA Annual Symposium Proceedings*, 2021.

N. Rehmat, **H. Farooq***, S. Kumar, S. ul Hussain, H. Naveed. "Predicting the pathogenicity of protein-coding mutations using Natural Language Processing." *IEEE EMBC*, pp. 5842–5846, 2020.

H. Farooq, H. Naveed. "Gpadrlex: Grouped phrasal adverse drug reaction lexicon." *IEEE ICSPCC*, pp. 1–6, 2019.

* Co-first author

Selected Research Projects

Predicting Target Genes of Non Coding Regulatory Variants Rχ | *Multi modal modeling, Hi C, eQTL, GWAS*

- Developed a scalable multi modal modeling framework integrating Hi C, eQTL, and GWAS data to predict causal regulatory variants and their target genes, validated against CRISPR perturbation datasets.

- Designed data integration and inference pipelines for large scale genomics datasets, enabling systematic mapping from noncoding variants to gene level effects.

Variant Driven Regulatory Interaction Modeling 39891345 | *Single cell modeling, Hi C, eQTL*
- Built computational models integrating chromatin interaction and genetic variation data to infer regulatory interactions governing gene expression at single cell resolution.
- Enabled mechanistic interpretation of variant effects by linking genome structure, regulatory activity, and expression variability.

Impact of LMNA Variant on Regulatory Architecture and AF Risk medRxiv | *Multi omics integration, population scale modeling*
- Integrated RNA seq, ATAC seq, and variant level regulatory features with large scale biobank data (All of Us) to model disease risk and regulatory mechanisms.
- Combined rare variant effects with polygenic risk modeling to connect chromatin state changes with complex disease phenotypes.

Enhancer Gene Target Prediction | *Machine learning, ATAC seq, ChIP seq, Hi C*
- Developed machine learning models integrating chromatin accessibility, histone marks, and 3D genome contacts to predict enhancer gene interactions across cell types.
- Implemented feature engineering and model evaluation pipelines for large scale regulatory genomics datasets.

Modeling Lamina Chromatin Interactions in Genome Architecture 40027763 | *3D genome modeling*
- Modeled the role of lamina associated chromatin in large scale genome organization and transcriptional regulation using multi omics data.

Hackathons

ALCF INCITE GPU Hackathon *Argonne National Laboratory (Lemont, IL), May 2025*
- Ported FoldRec C++ code from OpenACC to OpenMP offload for Intel GPUs on Aurora supercomputer.
- Integrated Parsl for scalable task management; optimized I/O and profiled performance using DAOS and `unitrace`.

ALCF INCITE GPU Hackathon *Argonne National Laboratory (Lemont, IL), May 2024*
- Ported FoldRec C++ code from CPU to OpenACC offload for NVIDIA GPUs on Polaris & Theta supercomputer.
- Achieved 10x speedup by parallelizing key bottleneck loops and profiling with Nsight Systems and HPCToolkit.

Honors & Awards

Fellowship: Malone Postdoctoral Fellow – Johns Hopkins University (2026)
Travel Awards: CSHL Epigenetics & Chromatin (2024); BPS Annual Meeting (2023); UIC Graduate Student Council (2023, 2024)
Presentation Awards: 1st Place Oral – UIC CBQB Research Day (2022); 1st Place Poster – UI Cancer Center Symposium (2022)
Academic Recognition: Dean's Honor List (4x); Undergraduate Research Grant – NGIRI, Pakistan (2017–18)

Selected Presentations

Biophysical Society (BPS) Annual Meeting: Oral and Poster – 2025 (Los Angeles); Poster – 2023 (San Diego), 2022 (San Francisco), 2021 (Virtual)
American Society of Human Genetics (ASHG): Poster – 2025 (Boston) , Poster – 2024 (Denver)
Cold Spring Harbor Laboratory (CSHL): Poster – Epigenetics & Chromatin (2024), Genome Organization & Nuclear Function (2022)
American Medical Informatics Association: Oral – Summit 2024 (Boston); Annual Symposium 2020 (Virtual)
UIC CBQB Research Day: Oral – 2024 (4th Annual), 2022 (2nd Annual), 2021 (1st Annual)
Other Conferences: Poster – UI Cancer Center Gary Kruh Symposium (2022, 2025); Oral – UIC BME Research Symposium (2025)

Peer Review Service

American Medical Informatics Association (AMIA): Annual Symposium (2022, 2025, 2026), Clinical Informatics Conference (2025), Amplify Informatics Conference (2025,2026)
IEEE Engineering in Medicine and Biology Society (EMBC): Annual Conference (2021, 2025, 2026)
Machine Learning in Computational Biology (MLCB) : Annual Conference (2026)